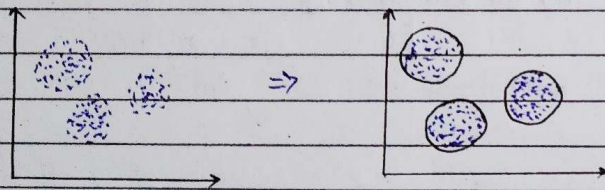


* UNIT-II → UNSUPERVISED LEARNING *

• Clustering →

- It is a type of USL algorithm.
- clustering is the task of dividing the population and data points into a number of groups such that data points in the same group are more similar to other data points in the same group and ~~also~~ dissimilar to the data points in other group.
- It is a collection of objects on the basis of similarity and dissimilarity between them.
- E.g. → The data points in the graph clustered together can be classified into single group we can distinguish the cluster and we can identify that there are three clusters in the graph.



• Applications of clustering →

- 1) Marketing → used to characterize and discover
- 2) ~~Biology~~ customer segments for marketing purpose.
- 3) ~~Library~~
- 2) Biology → it can be used for classification among different species of plants and animal.

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3) Library → used in clustering different books on the basis of topics and information.

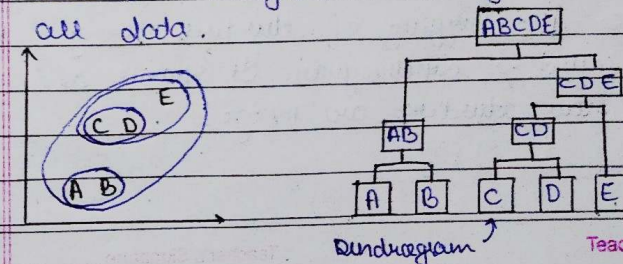
4) Insurance → used to acknowledge the customers, their policies and identifying the friends.

5) City Planning → used to make groups of houses and to study their values based on their geographical locations and other factors present.

6) Earthquake studies → by learning the earthquake, affected areas we can determine the dangerous zones.

• Hierarchical Agglomerative clustering →

- Also known as bottom-up approach or HAC.
- unlike flat clustering hierarchical clustering provides a structured way to group data.
- This clustering algorithm does not require us to pre-specify the number of clusters.
- Bottom-up algorithm treat each data as a single tone cluster at the outset and then successively agglomerate new pairs of clusters until all clusters have been merged into a single cluster that contains all data.



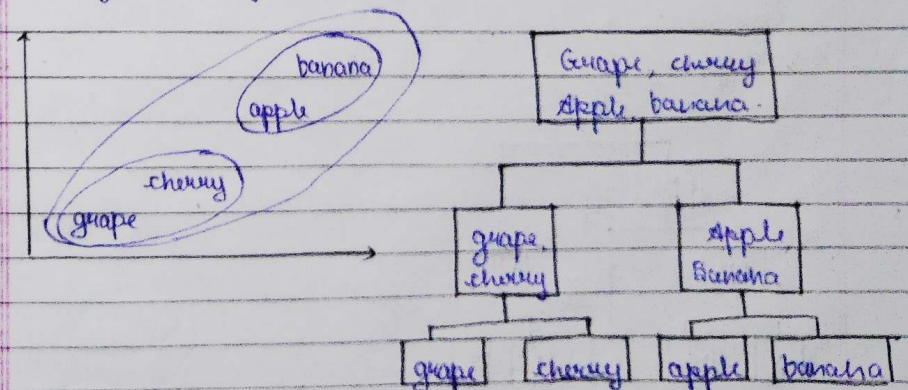
Dendrogram

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- workflow Hierarchical Agglomerative clustering ⇒
 - 1) start with individual points: each data point is its own cluster. for e.g. if you have 5 data points you start with 5 clusters each containing one ^{data} point.
 - 2) calculate distance between clusters: calculate distance between every pair of clusters. Initially since each cluster has one point this is the distance between two data points.
 - 3) Merge the closest cluster: identify two clusters with the smallest distance and merge them into a single cluster.
 - 4) Update Distance Matrix: after merging you now have one less cluster. Recalculate the distances between the new cluster and the remaining clusters.
 - 5) Repeat step 3 and step 4: keep merging the closest cluster and updating the distance matrix until you have only one cluster left.
 - 6) Create a Dendrogram: as the process continues you can visualize the merging of clusters using a tree like diagram called a dendrogram. It shows the hierarchy of how clusters are merged.

• Hierarchical clustering ⇒

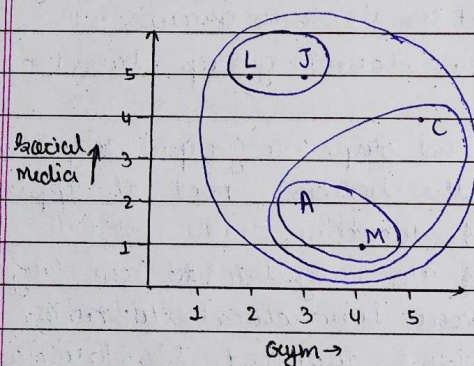
- It is used to group similar data points together based on their similarity creating a hierarchy or tree like structure.
- The key idea is to begin with each data point as its own separate cluster and then progressively merge or split them based on their similarity.
- e.g. ⇒ imagine you have 4 fruits with different weight apple (100gm), banana (120 gm), cherry (50gm), grape (30 gm). Hierarchical clustering starts by treating each fruit as its own group.
- It then merge the closest group based on their weight.
- First the cherry and grape are grouped together because they are the lightest, next the apple and banana are grouped together.
- Finally all fruits are merged in to one large group showing how hierarchical clustering progressively combines the most similar datapoints.



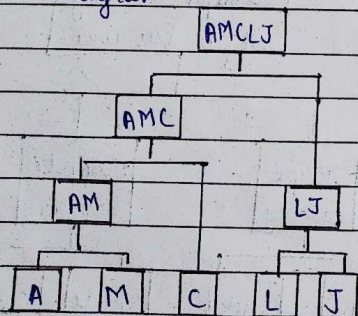
- Q. Draw hierarchical cluster and dendrogram on how many hours a week they spend on social media and gym.

	Social Media	Gym
Alan (A)	2	3
Lisa (L)	5	2
Joe (J)	5	3
Marc (M)	1	4
Carol (C)	4	5

⇒ For social media and Gym:



Dendrogram:



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K-Mean clustering ⇒

- It is an VSI algorithm which groups the unlabeled dataset into different clusters.
- K defines the number of predefined clusters that need to be created in the process as if $K=2$ there will be two clusters and for $K=3$ there will be three clusters and so on.
- It is an iterative algorithm that divides the unlabeled dataset into K different clusters in such a way that each dataset belongs to only one group that has similar properties.
- It is a centroid based algorithm where each cluster is associated with a centroid.
- The main aim of this algorithm is to minimize the sum of distances between the datapoints and their corresponding clusters.
- The algorithm takes the unlabeled dataset as input divides the dataset into K number of clusters and repeat the process until it does not find the best cluster.
- The value of K should be pre-determined in this algorithm.

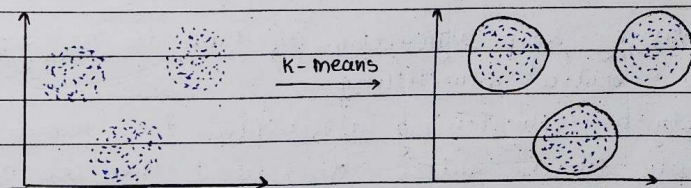


Figure before K-means cluster

Figure after K-means cluster

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• K-mean clustering Algorithm $\Rightarrow$

- 1) Select the number of K to decide the number of cluster.
- 2) Select random K points as centroids.
- 3) Assign each datapoint to their closest centroid, which will form the predefined K-clusters.
- 4) Calculate the variance or mean and place a new centroid of each cluster.
- 5) Repeat the 3rd step which means reassign each datapoint to the ~~new~~ new closest centroid of each cluster.
- 6) If any reassignment occurs go to step 4 else go to finish.
- 7) The model is ready.

Batchelor - Wilkin's Algorithm $\Rightarrow$

- $\rightarrow$ This algorithm is a method used in cluster analysis, particularly for determining the optimal number of clusters (K-value) in a dataset, which is often a challenge in algorithm like K-mean which as
- $\rightarrow$ It helps to solve the problem of selecting the best number of natural grouping in the data.

The step by step algorithm of BWA is as follows:

Step 1 $\Rightarrow$ Initial clustering.

- $\rightarrow$ Apply a clustering algorithm (e.g. K-mean) multiple times with different values of K.
- For e.g. $\rightarrow$ K=1, K=2, K=3, K=4

$\rightarrow$ This step groups the datapoints into clusters based on their similarity.

Step 2: Evaluate clustering quality

$\rightarrow$ For each K value, measure how good the clustering is using cluster validity indices such as:

(i) Silhouette Score $\Rightarrow$ measures how well separated clusters are.

(ii) Davies - Bouldin Index $\Rightarrow$ measures cluster similarity.

(iii) Calinski Harabasz Index $\Rightarrow$ measures variance between clusters.

$\rightarrow$ A good clustering means:

- 1) Points inside the same cluster are closed together (compactness).
- 2) Different clusters are far apart (good separation).

Step 3: Choose the best K.

$\rightarrow$ Compare the evaluation scores for all K values.

$\rightarrow$ Select the K that gives the best score.

$\rightarrow$ This represent the optimal number of clusters.

Step 4 $\Rightarrow$ Identify cluster centers

- $\rightarrow$ with the chosen K , determine the centroids (center points) of each cluster.
- $\rightarrow$ These represent the main center or mean point of the cluster.

Step 5 $\Rightarrow$ Final clustering

- $\rightarrow$ Assign every data point to the nearest centroid based on the chosen K .
- $\rightarrow$ This results in the final, most accurate clustering of the dataset.